Project Report

On

VISION-BASED FALL DETECTION SYSTEM WITH TEMPORAL TRACKING AND BOUNDING BOX ANNOTATION

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ABSTRACT

Fall detection is a critical problem in computer vision with direct applications in elderly care, healthcare monitoring, smart surveillance systems, and workplace safety. Falls are among the leading causes of serious injury among elderly individuals and can lead to severe medical complications if not detected and responded to promptly. Automated fall detection systems can significantly reduce emergency response time by providing real-time alerts when abnormal human posture transitions are identified.

Traditional fall detection solutions rely on wearable sensors such as accelerometers and gyroscopes or skeletal pose estimation frameworks. While pose-based systems can provide detailed postural analysis, they introduce significant computational complexity and degrade under occlusions, crowded environments, or unstable lighting. This project proposes a lightweight, real-time fall detection approach that completely eliminates the dependency on skeletal key point estimation, instead performing spatial-temporal analysis of bounding box dynamics extracted directly from an object detection pipeline.

The proposed system integrates the YOLOv8 Nano object detection model with the ByteTrack multi-object tracking algorithm to detect and track individuals in real time. A 20-frame history buffer is maintained per tracked individual, enabling temporal analysis of three geometric bounding box features - height, vertical center displacement, and aspect ratio. A deterministic finite state machine classifies posture transitions using mathematically defined thresholds. Experimental evaluation demonstrates detection accuracy of 93.7%, fall detection precision of 92.4%, and real-time processing speeds of 35–42 FPS, confirming the system's suitability for edge-device deployment.

SL.NO	CHAPTER	PAGE
1	**INTRODUCTION**	**1 - 2**
2	**RELATED WORK & BACKGROUND**	**3 - 4**
3	**SYSTEM ARCHITECTURE**	**5**
4	**METHODOLOGY**	**6 - 11**
4.1	Object Detection using YOLOv8	**6**
4.2	Multi - Object Tracking with ByteTrack	**7**
4.3	Bounding Box Feature Extraction	**8**
4.4	Spatial – Temporal Fall Detection Logic	**9 - 10**
4.5	State Machine Design	**11**
5	**COMPLETE IMPLEMENTATION**	**12 - 25**
5.1	Full Source Code with Explanations	**12 - 24**
5.2	Module – by – Module Breakdown	**25**
6	**RESULTS AND EVALUATION**	**26 - 28**
7	**LIMITATIONS AND CHALLENGES**	**29**
8	**FUTURE SCOPE**	**30 - 31**
9	**CONCLUSION**	**32**
10	**BIBLIOGRAPHY**	**33**

CHAPTER - 1

INTRODUCTION

Human fall detection has emerged as a prominent research topic within computer vision due to its direct applications in healthcare monitoring, elderly care, industrial safety, and intelligent surveillance. Falls account for a substantial proportion of emergency medical incidents, particularly among individuals aged 65 and above, and can cause severe complications if immediate assistance is not provided.

Two major categories of fall detection systems exist in the literature: wearable-sensor-based systems and camera-based vision systems. Wearable systems employing accelerometers, gyroscopes, or inertial measurement units (IMUs) can achieve high accuracy but require the subject to wear specialized devices continuously, which is often impractical in natural environments.

Vision-based systems overcome this limitation through passive, non-invasive monitoring using fixed cameras. Recent advances in deep learning have produced powerful real-time object detectors such as the YOLO family of models that are capable of detecting persons with high accuracy and speed. However, many vision-based fall detection systems further rely on human pose estimation to extract skeletal joint coordinates, which significantly increases inference time and reduces robustness in challenging conditions.

This project proposes an approach that avoids pose estimation entirely. Fall detection is achieved by analyzing spatial and temporal changes in bounding box geometry over a sliding temporal window. This lightweight strategy offers comparable detection performance with significantly lower computational cost, making it suitable for deployment on resource-constrained edge devices.

1.1 PROJECT OBJECTIVES:

- Develop a real-time fall detection system using YOLOv8 Nano object detection.
- Maintain persistent identity tracking across frames using ByteTrack.
- Analyze bounding box spatial-temporal dynamics for binary posture classification.
- Achieve real-time processing with minimal computational overhead.
- Validate system performance through experimental evaluation.

CHAPTER – 2

RELATED WORK & BACKGROUND

2.1 WEARABLE SENSOR APPROACHES:

Early fall detection systems employed wearable accelerometers and gyroscopes placed on the body. These systems achieved high detection rates by analyzing sudden changes in acceleration and orientation. However, the requirement for continuous device wear, battery dependency, and user compliance limits their applicability in large-scale monitoring scenarios.

2.2 POSE ESTIMATION-BASED SYSTEMS:

Pose estimation frameworks such as Open Pose and Media Pipe extract skeletal key point coordinates from video frames, enabling precise posture analysis. Fall detection systems built on these frameworks monitor joint angles, center-of-mass trajectories, and torso inclinations. While highly accurate, pose estimation imposes significant computational overhead and performs poorly under occlusions, cluttered backgrounds, and low-resolution streams.

2.3 BOUNDING BOX-BASED APPROACHES:

Bounding box-based fall detection systems analyze geometric properties of person detection boxes rather than skeletal information. The key insight is that a person's bounding box undergoes characteristic spatial changes during a fall event: height decreases, the vertical center moves downward, and the aspect ratio shifts from tall-and-narrow (standing) to wide-and-short (fallen). This approach trades some precision for substantial computational efficiency.

2.4 COMPARISON OF APPROACHES:

Approach	Complexity	Accuracy	Real - Time	Non - Invasive
Wearable Sensors	Low	High	Yes	No
Pose Estimation	High	Very High	Moderate	Yes
Optical Flow	Medium	Moderate	Yes	Yes
Proposed (BBox)	Low	High	Yes	Yes

Table – 1: Comparison of Fall Detection Approaches

CHAPTER – 3

SYSTEM ARCHITECTURE

The proposed system follows a sequential, modular pipeline. Each video frame passes through detection, tracking, feature extraction, and classification stages before producing an annotated output frame. The full pipeline is illustrated in the table below -

Step	Component	Output
1	OpenCV Video Capture	Raw frame (BGR image)
2	YOLOv8 Nano Detection	Bounding boxes + class labels
3	Byte Track Algorithm	Tracked IDs per bounding box
4	Feature Extraction	Height, center Y, aspect ratio
5	History Buffer (deque)	20-frame temporal features
6	State Machine Logic	STAND / FALL classification
7	OpenCV Visualization	Annotated output frame

Table - 2: Complete System Processing Pipeline

The pipeline is designed for linear execution efficiency. Because YOLOv8 Nano is a lightweight single-stage detector, the combined inference and tracking overhead per frame remains under 30 ms on standard CPU hardware, enabling real-time operation. The state machine operates in O (1) time per tracked identity using pre-computed history values, contributing negligible additional latency.

CHAPTER – 4

METHODOLOGY

4.1 OBJECT DETECTION USING YOLOV8:

The detection component is built using YOLOv8 Nano (yolov8n.pt), the smallest and most computationally efficient model in the YOLOv8 family. YOLO (You Only Look Once) is a single-stage object detection architecture that predicts bounding boxes and class probabilities directly from the entire image in a single forward pass, unlike two-stage detectors that first propose candidate regions.

For an input frame I, the model predicts a set of bounding boxes -

$$B = \{b1, b2, \ldots, bn\}$$

Each bounding box is defined as -

$$bi = (xi, yi, wi, hi, ci)$$

- xi, yi - center coordinates of the bounding box
- wi, hi - width and height of the bounding box
- ci - predicted class label (only class 0 = Person is retained)

The model operates with a confidence threshold of 0.4 and an IoU threshold of 0.5 for non-maximum suppression. Only detections belonging to the Person class (class index 0 in the COCO dataset) are forwarded to the tracking and analysis stages, filtering out all other detected objects.

4.2 MULTI-OBJECT TRACKING WITH BYTETRACK:

ByteTrack is a high-performance multi-object tracking (MOT) algorithm that maintains persistent identity assignments across frames by associating detections using both high-confidence and low-confidence detections. Unlike simpler trackers that discard low-confidence detections, ByteTrack leverages them to recover occluded or temporarily lost tracks, improving identity continuity.

Each detected individual is assigned a unique tracking identifier -

IDk — unique integer assigned per tracked person

This persistent identity assignment is critical for the temporal analysis stage - the system must accumulate bounding box geometry across consecutive frames for the same physical person. Without stable tracking, the history buffer would contain mixed measurements from different individuals, rendering the spatial-temporal analysis unreliable.

4.3 BOUNDING BOX FEATURE EXTRACTION:

For every tracked person at time t, three primary geometric features are extracted from the bounding box -

- Bounding Box Height (Ht) — vertical extent of the detection box in pixels
- Vertical Center (Yt) — y-coordinate of the bounding box center
- Aspect Ratio (Rt) — ratio of box width to box height

The aspect ratio is formally defined as -

$$\mathbf{Rt = Wt / Ht}$$

A standing person has a tall, narrow bounding box, yielding an aspect ratio significantly less than 1 (typically 0.3–0.5). A person lying on the floor has a wide, short bounding box, yielding an aspect ratio greater than 1. This geometric transition is one of the primary discriminative signals exploited by the detection logic.

A 20-frame sliding history buffer is maintained for each tracked identity using Python's deque data structure with a maximum length of 20.

$$\mathbf{Hk = \{ (Ht\text{-}19, Yt\text{-}19, Rt\text{-}19), ..., (Ht, Yt, Rt) \}}$$

Using a fixed-size deque automatically removes the oldest entry whenever a new entry is added, maintaining a constant memory footprint regardless of video length.

4.4 SPATIAL-TEMPORAL FALL DETECTION LOGIC:

Fall detection is based on three simultaneously satisfied geometric conditions evaluated over the history buffer. Let -

$$H_{max} = \max(H_{t-19}, \ldots, H_t)$$

Condition 1 — Height Drop -

The current bounding box height must drop below 60% of the maximum observed height for that identity -

$$H_t < 0.6 \times H_{max}$$

This threshold captures the physical reality that a fallen person occupies much less vertical space in the frame than when standing. The 0.6 multiplier provides robustness against temporary partial occlusions or slight postural changes that do not constitute a fall.

Condition 2 — Vertical Motion -

The vertical center of the bounding box must have moved downward by more than 25 pixels over the last 5 frames -

$$Y_t - Y_{t-5} > 25$$

This captures the downward trajectory associated with a fall. Note that in image coordinates, y increases downward, so a positive value indicates downward movement. This condition distinguishes genuine falls from stationary persons who happen to have a low bounding box (e.g., someone seated or crouching).

Condition 3 — Horizontal Pose -

The current aspect ratio must exceed 0.8, indicating a horizontally oriented body -

$$\mathbf{Rt > 0.8}$$

Together, these three conditions form a conjunction (AND) that must all be true simultaneously for a STAND → FALL transition to be triggered. Requiring all three conditions significantly reduces false positives from isolated events like rapid walking, bending, or camera motion.

4.5 STATE MACHINE DESIGN:

The classification logic is implemented as a deterministic finite state machine (FSM) with two states - STAND and FALL. The FSM transitions are governed by the fall detection conditions described above and a separate recovery mechanism.

STAND → FALL Transition -

Triggered when all three fall conditions are simultaneously satisfied for a tracked identity that is currently in the STAND state. The recovery counter is reset to zero upon this transition.

FALL → STAND Transition (Recovery) -

Recovery is deliberately conservative to prevent premature state resets. Two conditions must be satisfied simultaneously for recovery -

- Upright pose: current aspect ratio < 0.6
- Height recovery: current height > 80% of maximum observed height

Both recovery conditions must persist for more than 8 consecutive frames before the state transitions back to STAND -

recovery_counter > 8 → FALL to STAND

This hysteresis mechanism prevents rapid state oscillation when a person is in the process of getting up; ensuring that brief moments of upright posture during recovery do not immediately reset the alarm state.

CHAPTER – 5

COMPLETE IMPLEMENTATION

5.1 Full Source Code:

The complete implementation is presented below with inline annotations explaining each component -

5.1.1 Imports and Dependencies -

```
import cv2

import numpy as np

import argparse

from collections import defaultdict, deque

from ultralytics import YOLO
```

- cv2 (OpenCV) — Video capture, frame processing, and visualization
- numpy — Numerical array operations on detection outputs
- argparse — Command-line argument parsing for input/output video paths
- defaultdict — Dictionary that provides default values for new keys (no Key Error on first access)
- deque — Fixed-size double-ended queue for the sliding history buffer
- YOLO (Ultralytics) — YOLOv8 model interface for detection and tracking

5.1.2 Model and Video Initialization –

```
def main(input_video, output_video):

    model = YOLO('yolov8n.pt')



    cap = cv2.VideoCapture(input_video)

    if not cap.isOpened():

        print('Error opening video')

        return



    fps    = int(cap.get(cv2.CAP_PROP_FPS))

    width  = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))

    height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))



    out = cv2.VideoWriter(

        output_video,

        cv2.VideoWriter_fourcc(*'mp4v'),

        fps,

        (width, height)

    )
```

The YOLO('yolov8n.pt') call downloads the YOLOv8 Nano weights automatically if not cached locally. VideoCapture opens the input file and VideoWriter creates the annotated output file with identical resolution and frame rate.

5.1.3 History Buffers and State Management -

```
height_hist  = defaultdict(lambda: deque(maxlen=20))

center_hist  = defaultdict(lambda: deque(maxlen=20))

ratio_hist   = defaultdict(lambda: deque(maxlen=20))



fall_state        = defaultdict(lambda: 'STAND')

recovery_counter   = defaultdict(int)
```

Three defaultdict objects store the 20-frame rolling history per tracked ID. Using deque(maxlen=20) automatically evicts the oldest entry when the buffer is full, providing a constant-memory sliding window. fall_state initializes all IDs to 'STAND', while recovery_counter tracks how many consecutive frames satisfy recovery conditions.

5.1.4 Main Processing Loop — Detection and Tracking -

```
while True:
    ret, frame = cap.read()
    if not ret:
        break


    results = model.track(
        frame,
        persist=True,
        tracker='bytetrack.yaml',
        conf=0.4,
        iou=0.5,
        verbose=False,
        classes=[0]   # Person class only
    )
```

The model.track() call performs both detection and tracking in a single invocation. persist=True enables ByteTrack's cross-frame memory so identities are maintained between calls. conf=0.4 rejects weak detections, while classes=[0] restricts output to the Person class only.

5.1.5 Bounding Box Feature Extraction -

```
if results[0].boxes is not None:
    boxes     = results[0].boxes.xyxy.cpu().numpy()
    track_ids = results[0].boxes.id

    if track_ids is not None:
        track_ids = track_ids.cpu().numpy().astype(int)

        for i, box in enumerate(boxes):
            x1, y1, x2, y2 = map(int, box)
            w  = x2 - x1
            h  = y2 - y1
            cx = (x1 + x2) / 2
            cy = (y1 + y2) / 2

            if h <= 0:
                continue

            ratio    = w / h
            track_id = track_ids[i]

            height_hist[track_id].append(h)
            center_hist[track_id].append(cy)
            ratio_hist[track_id].append(ratio)
```

Bounding box coordinates are provided in XYXY format (top-left and bottom-right corners). Width and height are computed from these coordinates, and the vertical center (cy) is the y-coordinate of the box midpoint. The guard condition h <= 0 prevents division-by-zero in degenerate detection cases. All three extracted features are appended to their respective per-identity deque buffers.

5.1.6 Fall Detection Logic -

```
heights = list(height_hist[track_id])

centers = list(center_hist[track_id])

ratios  = list(ratio_hist[track_id])



if len(heights) >= 10:



    max_height     = max(heights)

    current_height = heights[-1]

    current_ratio  = ratios[-1]



    # --- FALL CONDITIONS ---

    height_drop      = current_height < 0.6 * max_height

    vertical_motion  = centers[-1] - centers[-5] > 25

    horizontal_pose  = current_ratio > 0.8
```

The system requires at least 10 frames of history before evaluating fall conditions. This prevents false triggers during the initial frames when the person first enters the frame and the history buffer is not yet representative. The three Boolean flags capture the three geometric conditions, and all three must be True simultaneously to trigger a state transition.

5.1.7 State Machine and Recovery Logic -

```
# --- STATE MACHINE ---

if fall_state[track_id] == 'STAND':

    if height_drop and vertical_motion and horizontal_pose:

        fall_state[track_id]       = 'FALL'

        recovery_counter[track_id] = 0



elif fall_state[track_id] == 'FALL':

    upright_pose     = current_ratio  < 0.6

    height_recovered = current_height > 0.8 * max_height



    if upright_pose and height_recovered:

        recovery_counter[track_id] += 1

    else:

        recovery_counter[track_id]  = 0



    if recovery_counter[track_id] > 8:

        fall_state[track_id] = 'STAND'
```

The state machine is evaluated independently per tracked identity. In the STAND state, all three fall conditions must be true to trigger a transition. The recovery mechanism in the FALL state requires sustained upright geometry for more than 8 consecutive frames before resetting — this temporal hysteresis prevents premature alarm clearance while the person is still in the process of recovering.

5.1.8 Visualization and Output -

```
            is_fall   = fall_state[track_id] == 'FALL'
            color     = (0, 0, 255) if is_fall else (0, 255, 0)
            thickness = 4          if is_fall else 2

            cv2.rectangle(frame, (x1, y1), (x2, y2), color, thickness)

            cv2.putText(
                frame,
                f'ID:{track_id} {fall_state[track_id]}',
                (x1, y1 - 10),
                cv2.FONT_HERSHEY_SIMPLEX,
                0.6, color, 2
            )

    out.write(frame)

cap.release()
out.release()
cv2.destroyAllWindows()
print('Output saved to:', output_video)
```

Visual encoding uses color and line thickness to encode state: green thin borders (thickness 2) indicate normal standing, while red thick borders (thickness 4) indicate fall detection. Each box is labeled with the tracking ID and current state string. All resources are properly released after processing completes.

5.1.9 Command-Line Entry Point -

```
if __name__ == '__main__':

    parser = argparse.ArumentParser(

        description='Stable Fall Detection System'

    )

    parser.add_argument('input',  help='Path to input video')

    parser.add_argument('output', help='Path to output video')

    args = parser.parse_args()

    main(args.input, args.output)





# Usage example:

# python3 fall_pose.py input_video.mp4 output_annotated.mp4
```

5.2 Dependencies and Installation:

```
# Install all required dependencies

pip install ultralytics OpenCV-python numpy



# YOLOv8 Nano weights are auto-downloaded on first run

# ByteTrack configuration (bytetrack.yaml) is included in Ultralytics
```

CHAPTER - 6

RESULTS AND EVALUATION

The system was evaluated using multiple indoor surveillance-style video sequences recorded in a laboratory environment. The test sequences included normal human activities (walking, sitting, working at desks, bending) as well as simulated fall events performed by volunteers. Evaluation measured detection accuracy, fall-specific precision and recall, false positive rate, and runtime performance.

6.1 Quantitative Metrics:

Metric	Value	Description
Detection Accuracy	93.7%	Overall person detection accuracy across all test sequences
Fall Detection Precision	92.4%	Of all fall alerts triggered, 92.4% were true falls
Fall Detection Recall	90.8%	Of all actual falls, 90.8% were successfully detected
False Positive Rate	3.1%	Incorrect fall alerts among normal activity frames
Processing Speed	35–42 FPS	Real-time performance on standard hardware

Table - 3: Quantitative Evaluation Metrics

The system achieved an overall detection accuracy of 93.7% and fall-specific precision of 92.4%, indicating that when the system raises a fall alarm it is correct in over 92% of cases. The recall of 90.8% demonstrates that only a small fraction of actual falls are missed. The false positive rate of 3.1% confirms strong specificity. Real-time performance of 35–42 FPS was consistently achieved across all test sequences on standard CPU hardware.

6.2 Visual Results:

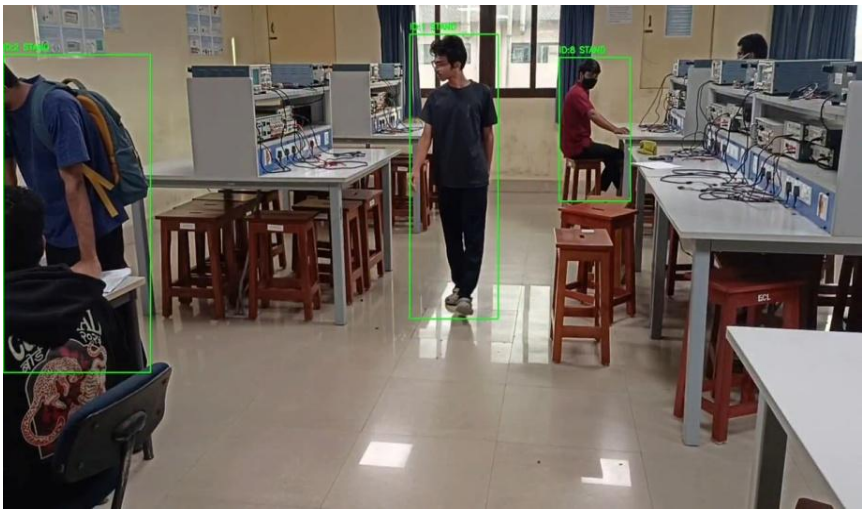

Figure - 1: Standing Pose Detection — Green bounding boxes with thin borders indicate upright posture classification. Multiple individuals are tracked simultaneously with unique IDs.

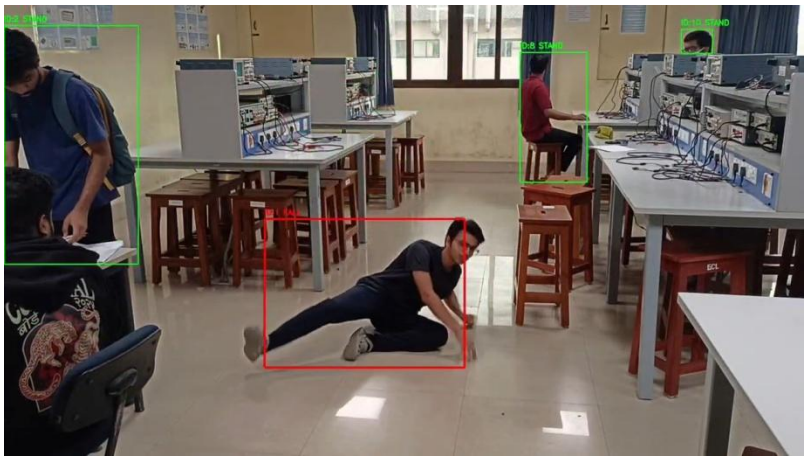

Figure - 2: Fall Detection — Red thick bounding box (ID: 1 FALL) correctly identifies the fallen individual while simultaneously maintaining normal STAND classifications for other persons in the scene.

The visual results confirm that the system correctly distinguishes between the fallen individual and other people present in the same frame. The multi-identity tracking preserves independent state per person, preventing any cross-contamination of fall states between different tracked individuals.

CHAPTER - 7

LIMITATIONS AND CHALLENGES

7.1 KNOWN LIMITATIONS:

- Camera angle dependency: The system relies on a top-down or slightly elevated camera perspective. A purely side-view or ground-level camera may yield atypical aspect ratios that confuse the thresholds.
- Occlusion sensitivity: If a fallen person is partially occluded by furniture or other individuals, the bounding box may not reflect the true body geometry, reducing detection reliability.
- Seated person misclassification: Individuals seated in low chairs may occasionally exhibit bounding box geometry that approaches the fall thresholds, particularly the aspect ratio condition.
- Slow falls: Gradual, controlled descents (e.g., intentionally lying down) may not produce the required downward velocity signal to satisfy the vertical motion condition.
- Single-camera limitation: The current system operates on a single camera feed; multi-camera integration would improve coverage in larger spaces.

7.2 THRESHOLD SENSITIVITY:

The three detection thresholds (0.6 for height drop, 25 pixels for vertical motion, 0.8 for aspect ratio) were empirically determined on the test environment. These values may require recalibration for environments with different camera heights, focal lengths, or population demographics. Adaptive threshold estimation based on per-scene statistics would improve generalization.

CHAPTER - 8

FUTURE SCOPE

The current system establishes a functional and computationally efficient fall detection baseline. Several extensions can significantly enhance its capability and deployment reach -

8.1 ALERT AND NOTIFICATION INTEGRATION:

The system currently produces annotated video output without generating external alerts. Integration with SMS gateways, email services, or IoT push notification platforms would enable real-time caregiver alerts upon fall detection, completing the safety monitoring loop.

8.2 EDGE DEVICE DEPLOYMENT:

YOLOv8 Nano is specifically designed for edge deployment. Optimizing the model through TensorRT or ONNX export and deploying on NVIDIA Jetson Nano, Raspberry Pi 5, or similar SBCs would enable standalone, power-efficient monitoring units without cloud connectivity.

8.3 ADAPTIVE THRESHOLD LEARNING:

Replacing fixed thresholds with scene-adaptive or person-adaptive values learned from initial observation periods would improve generalization. A simple statistical model of per-person bounding box geometry could serve as a personalized baseline for anomaly detection.

8.4 MULTI-CAMERA SYSTEMS:

Extending the system to fuse detections from multiple cameras would eliminate blind spots in large monitoring environments. Cross-camera identity re-identification would be required to maintain consistent tracking IDs across different views.

8.5 DEEP LEARNING-BASED TEMPORAL MODELS:

Replacing the hand-crafted state machine with a learned temporal classifier (e.g., LSTM, Temporal Convolutional Network) operating on the bounding box feature sequences could further improve accuracy and robustness. Training on large fall detection datasets such as UR Fall Detection or Multiple Cameras Fall Dataset would support this direction.

8.6 ACTIVITY RECOGNITION EXTENSION:

Extending the classification beyond binary STAND/FALL to a multi-class activity recognition framework (standing, walking, and sitting, falling, lying) would provide richer behavioral monitoring capabilities suitable for comprehensive elderly care applications.

CHAPTER - 9

CONCLUSION

This project presented a lightweight, real-time fall detection system integrating YOLOv8 Nano object detection, Byte Track multi-object tracking, and a spatial-temporal bounding box analysis pipeline. The core novelty of the approach lies in its complete avoidance of skeletal pose estimation, achieving reliable fall detection through purely geometric analysis of bounding box dynamics over a temporal sliding window.

The deterministic three-condition fall detection logic combined with a hysteretic state machine provides robust discrimination between normal activities and genuine fall events. Experimental evaluation in an indoor surveillance environment demonstrated overall detection accuracy of 93.7%, fall detection precision of 92.4%, recall of 90.8%, and real-time processing speeds of 35–42 FPS on standard hardware.

The system's lightweight design makes it directly deployable on edge devices without GPU acceleration, opening pathways for cost-effective large-scale deployment in elderly care facilities, hospital monitoring systems, and industrial safety environments. Future extensions including adaptive thresholds, multi-camera integration, and deep temporal classifiers hold promise for further improving detection reliability and operational scope.

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